# Hyperspectral Pansharpening Using Deep Prior and Dual Attention Residual Network

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***Abstract*— Convolutional neural networks (CNNs) have recently achieved impressive improvements on hyperspectral (HS) pansharpening. However, most of the CNN-based HS pansharpening approaches would have to first upsample the low-resolution hyperspectral image (LR-HSI) using bicubic interpolation or data-driven training strategy, which inevitably lose some details or greatly rely on the learning process. In addition, most previous methods regard the pansharpening as a black-box problem and treat diverse features equally, thus hindering the discriminative ability of CNNs. To conquer these issues, a novel HS pansharpening method using deep hyperspectral prior (DHP) and dual-attention residual network (DARN) is proposed in this article. Specifically, we first upsample the LR-HSI to the scale of the panchromatic (PAN) image through the DHP algorithm, which can better preserve spatial and spectral information without learning from large data sets. The upsampled result is then concatenated with the PAN image to form the input of the DARN, where several channel-spatial attention residual blocks (CSA ResBlocks) are stacked to map the residual HSI between the reference HSI and the upsampled HSI. In each CSA ResBlock, two complementary attention modules, i.e., channel attention and spatial attention modules, are designed to adaptively learn more informative features of spectral channels and spatial locations simultaneously, which can effectively boost the fusion accuracy. Finally, the fused HSI is obtained by the summation of the upsampled HSI and the reconstructed residual HSI. The experimental results of both simulated and real HS data sets demonstrate that the performance of our DHP-DARN method is superior over the state-of-the-art HS pansharpening approaches.**

***Index Terms*— Deep hyperspectral prior (DHP), dual-attention residual network (DARN), hyperspectral (HS) pansharpening.**

## I. Introduction

HYPERSPECTRAL images (HSIs) with rich spectral characteristics across a large number of wavelengths have gained momentum in the field of remote sensing and shown significant performance improvements in wide application domains, such as scene classification, anomaly detection, and spectral unmixing [1]–[3]. However, due to the hardware limitations in the design of remote sensing imaging system, there is the inevitable tradeoff among spatial resolution, spectral resolution, and signal-to-noise ratio (SNR) [4]. Specifically, the hyperspectral (HS) imaging sensors can simultaneously capture numerous bands within a narrow spectral window and, thus, only provide relatively low spatial resolution to satisfy the sufficient SNR [5], which restricts the range of potential applications of HSIs. In contrast, the panchromatic (PAN) image with a single band acquired by the PAN imaging system usually has much higher spatial resolution. Therefore, an efficient way to generate a high-resolution HSI (HR-HSI) is fusing a low-resolution HSI (LR-HSI) with a high-resolution PAN (HR-PAN) image of the same scene. This procedure is called as HS pansharpening, which is an extension to HSIs of the conventional pansharpening referring to the fusion of a multispectral image (MSI) and a PAN image [6]. The technology of HS pansharpening is able to enhance the spatial resolution of HSIs effectively, which greatly contributes to the accurate identification and classification of the observed materials and scenes [7].

In recent decades, various traditional HS pansharpening techniques have been proposed and developed, which can roughly be categorized into five major branches: component substitution (CS), multiresolution analysis (MRA), hybrid, Bayesian, and matrix factorization methods [8]. Algorithms of the first category rely on substituting the spatial component of the HSI with the PAN image, which have been provided in most of the professional remote sensing software due to their fast and easy implementation. The widely employed algorithms such as Gram–Schmidt adaptive (GSA) [9] and principal component analysis (PCA) [10] are under the class

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of CS. MRA methods inject the spatial details extracted by a multiresolution decomposition of the PAN image into the resampled HSI, including algorithms such as smoothing filter-based intensity modulation (SFIM) [11] and MTF-GLP with high-pass modulation (MTF-GLP-HPM, abbreviated as MGH) [12]. In the former two branches, although the CS methods generally generate fused results with sharp spatial structures in most cases, they sometimes suffer from significant spectral distortions. In contrast, the MRA family performs better in spectral preservation, but it is more sensitive to registration errors, which may cause serious distortions in the spatial domain [13]. The guided filter PCA (GFPCA) [14] is one of the typical hybrid algorithms, which won the "Best Paper Challenge" Award of the IEEE GRSS Data Fusion Contest in 2014. The Bayesian methods formulate the fusion problem within the Bayesian inference framework, which provides a convenient way to regularize the fusion by utilizing an appropriate prior distribution. The Bayesian naive Gaussian prior (Bayesian naive, abbreviated as BNA) [15] and Bayesian sparsity promoted Gaussian prior (Bayesian sparse, abbreviated as BSP) [16] belong to this branch. As one of the representatives of the matrix factorization-based methods, the coupled nonnegative matrix factorization (CNMF) [17] approach regularizes the fusion problem by using the priors of spectral unmixing. The last two branches can also be categorized as the model-based method, which requires appropriate regularizations to build an energy function based on the prior knowledge or some reasonable assumptions of the fusion process to reconstruct the HR-HSI. However, the fusion performance is generally limited by the insufficient representation ability of employed models, and severe quality degradation may be caused in the cases where the assumptions do not fit the problem [18]. Furthermore, most existing optimization solutions of the fusion model are solved iteratively, which is time-consuming and low efficiency.

Recently, deep learning (DL) models, especially convolutional neural networks (CNNs), have achieved significant performance advantages in the domain of image spatial resolution enhancement, due to their strong capability to automatically learn high-level features from input data. Among them, Dong *et al.* [19] proposed a super-resolution CNN (SRCNN) by first introducing a shallow three-layer CNN for image super-resolution (SR). Based on the effective residual learning strategy [20], Ledig *et al.* [21] introduced the residual network (ResNet) to build a deeper network for image SR, which was further improved in EDSR and MDSR proposed by Lim *et al.* [22] using simplified residual blocks (ResBlocks). To reduce the computational complexity of direct super-resolving details, Lai *et al.* [23] designed the LapSRN, where the high-frequency residuals were progressively reconstructed in a coarse-to-fine manner. Although considerable improvement of image SR has been made, these existing CNN-based SR methods mainly focus on the depth or width of network design, which lacks the ability to exploit discriminative features across channels, thus resulting in relatively limited representational power of CNNs. With the recent success of attention mechanism in various vision tasks [24], it has been introduced to deep CNNs to obtain better improvements of image SR. By first considering the channel attention (CA) for image SR, Zhang *et al.* [25] presented a very deep residual channel attention network (RCAN). Compared with the RCAN, the spatial color attention network (SCAN) [26] achieved superior performance by jointly exploiting the spatial and spectral dependence within color images. Such an attention mechanism allows an SR network to focus on more useful features and enhance the discriminative learning capacity.

Motivated by the astonishing performance achieved by CNNs in improving image resolution, researchers have successfully applied CNNs to spatial enhancement of remote sensing data and presented several CNN-based remote sensing image fusion methods [27]–[34], which take a lead position in this field. However, most of these methods apply CNNs to remote sensing images in the same or similar ways as in RGB images, which regard the sharpening procedure as a black-box DL problem and treat the features extracted from input images equally, thus limiting the ability to selectively use informative features. To the best of our knowledge, only a few attempts in [35] and [36] have utilized the attention mechanism to super-resolve a single remotely sensed imagery with quite a few spectral bands and relatively high spatial resolution. When focusing on the resolution enhancement problem of the HSI with much wider spectral range and lower spatial resolution, some HR auxiliary data (e.g., the HR-PAN image) of the same scene are easier to obtain, but the fusion of an HSI and a PAN image in the DL framework is still faced with challenges, which mainly lie in the following three aspects.

1) HSIs usually have abundant spectral and spatial information, and thus, how to fully exploit and fuse the complementary information extracted from HSIs and PAN images remains an important issue. On the one hand, some adjacent bands of an HSI are highly correlated. On the other hand, the nonlocal similarity exists across spatial domain. For HS pansharpening, both spectral and spatial redundancies should be made full use of.
2) Many of the existing CNN-based pansharpening methods have to first upscale the LR-HSI to the size of the PAN image using bicubic interpolation or data-driven learning strategy, which inevitably lose some information details or greatly rely on the ability of knowledge adaptation.
3) Most recent CNN-based pansharpening models consider that the extracted features are equally important, which lacks flexibility in dealing with different types of information. For instance, the high-frequency information incorporating HR spatial details extracted from textured areas can be more difficult to sharpen than that of flat regions, where more attention should be paid to during the pansharpening process.

To tackle these challenges practically, we propose a novel framework based on deep hyperspectral prior (DHP) and dual-attention residual network (DARN) for HS pansharpening in this article. We consider four main contributions in this article as follows.

1) DHP captured by the structure of a U-Net is investigated for upsampling the LR-HSI effectively, which can obtain

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better spatial resolution enhancement and spectral information preservation without a large amount of example images for training.

2) A new DARN with several channel-spatial-attention residual blocks (CSA ResBlocks) is proposed. To the best of our knowledge, this is the first study that integrates the channel-spatial-attention mechanism to address the issue of HS pansharpening, which can adaptively learn more informative channelwise and spatial-domain features simultaneously.
3) We develop the CSA ResBlock that combines both the CA module and the spatial-attention (SA) module for HS pansharpening. The proposed CSA ResBlock is able to fully exploit interchannel and interspatial relationships of feature maps concurrently, which can further improve the pansharpening accuracy by motivating more relevant features and suppressing less useful ones of certain spectral channels and spatial positions.
4) The residual learning strategy is adopted to speed up the convergence and boost the fusion performance. Extensive experiments conducted on both simulated and real HS data sets demonstrate the superiority of our DHP-DARN method over the state-of-the-art HS pansharpening approaches.

The rest of this article is organized as follows. Section II introduces some basics of deep image prior (DIP) for SR and CNN-based remote sensing image fusion. In Section III, the proposed DHP-DARN method is described in detail. In Section IV, the experimental results and discussions of different data sets are presented. Conclusions are drawn in Section V.

## II. Related Works

### A. DIP for SR

Deep CNNs have led to notable improvements in the task of image SR, which is usually imputed to their ability to learn realistic image priors from a great quantity of example images. Contrary to the generally accepted belief that learning is necessary for building image priors, the recently proposed DIP method [37] has shown that a randomly initialized CNN can be used as an effective handcrafted prior for image SR with excellent results, which does not require a large data set for training.

As one of the image restoration problems reported in [37], the SR task with the goal of taking an LR image $y$ and an upsampling factor $\alpha$ to generate the target HR image $x$ can be formulated as follows:

$$x^* = \min_x E(x, y) + R(x) \tag{1}$$

where $E(x, y)$ is an energy function, $R(x)$ is a regularization term, and $x^*$ denotes the restored HR image. Different from the majority of CNN-based methods using the learned regularizers from images, the DIP approach replaces the $R(x)$ with the implicit prior captured by the network itself, which can be expressed as follows:

$$x^* = f_{\theta^*}(z), \quad \theta^* = \operatorname{argmin}_\theta E(f_\theta(z), y) \tag{2}$$

where $f_\theta$ represents the network with parameters $\theta$, and $z$ is the input tensor filled with uniform noise that has the same spatial size with $x$. Given the minimizer $\theta^*$, the SR result can be obtained as $x^* = f_{\theta^*}(z)$. Moreover, the $L_2$-distance between the downsampled result of the generated HR image $f_\theta(z)$ and the LR image $y$ is utilized as the energy function $E(\cdot)$ to constrain the output of the architecture

$$E(f_\theta(z), y) = \|d(f_\theta(z)) - y\|^2 \tag{3}$$

where $d(\cdot)$ denotes a downsampling operator, which resizes an image with a scale factor $\alpha$. Hence, the DIP method can seek the optimal parameters to generate an HR image which, when downsampled, is as close to the LR image $y$ as possible.

With the success of the DIP algorithm, Sidorov and Hardeberg [38] have proposed the DHP method for the task of HSI restoration, which is an effective extension to the HSI domain. To carry out the HS pansharpening procedure, most methods first interpolate the LR-HSI to the same size with the PAN image, which may introduce extra faulty information and, thus, degrade the fusion accuracy. The DHP approach that uses the structure of the network as the image prior can produce remarkable SR results without data-driven training, which is of great benefit to exploring the potential prior regularization for HS pansharpening due to the problem of limited HSI training sets.

### B. CNN-Based Remote Sensing Image Fusion

Over recent years, CNN-based methods have emerged rapidly in the field of remote sensing image fusion since CNNs can effectively characterize very complex nonlinear relationship between input and target data. Among these methods, Masi *et al.* [27] presented a simple three-layer pansharpening CNN (PNN) taking the upsampled MSI stacked with the associated PAN image as input, which was similar to the structure of SRCNN [19]. Inspired by the wide application of the ResNet [20] in image SR tasks, a deep residual pansharpening neural network (DRPNN) [28] was proposed to learn the residuals of the input and ground-truth MSIs. The PanNet for multispectral (MS) pansharpening was presented by Yang *et al.* [29], in which the mapping was learned in the high-pass filtering domain rather than the image domain. Considering the 3-D characteristics of HSIs, Palsson *et al.* [30] proposed a new method for the fusion of HS and MS images using a 3-D-CNN, which showed promising results when the HSI was corrupted by additive noise. Dian *et al.* [31] proposed a deep HSI sharpening (DHSIS) method, which incorporated the priors leaned by CNN into the fusion of LR-HSI and HR-MSI. To improve the spatial resolution of HSI, Li *et al.* [32] presented a detail-based deep Laplacian pansharpening (DDLPS) approach, which used the LapSRN [23] to upsample each band of the LR-HSI successively. In order to strengthen the spectral prediction capability of HS pansharpening networks, two spectrally predictive CNN models (named HyperPNN1 and HyperPNN2) were designed in [33]. For boosting the accuracy of HS pansharpening, Zheng *et al.* [34] proposed the deep residual CNN (DRCNN), which obtained significant performance improvements in edge details enhancement.

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TABLE I
SUMMARY OF CNN-BASED REMOTE SENSING IMAGE FUSION METHODS

| Name | Brief Descriptions | Specific Applications |
|---|---|---|
| PNN [27] | a three-layer pansharpening CNN | MS pansharpening |
| DRPNN [28] | a deep residual pansharpening CNN | |
| PanNet [29] | learning by a ResNet [20] in the high-pass domain | |
| 3D-CNN [30] | using a 3D-CNN to fuse together HS and MS images | HS-MS image fusion |
| DHSIS [31] | learning priors by deep CNN-based residual learning | |
| DDLPS [32] | a detail-based method using the LapSRN [23] for LR-HSI SR | HS pansharpening |
| HyperPNN [33] | strengthening the spectral prediction capability of CNN | |
| DRCNN [34] | a deep residual CNN for boosting the fusion accuracy | |

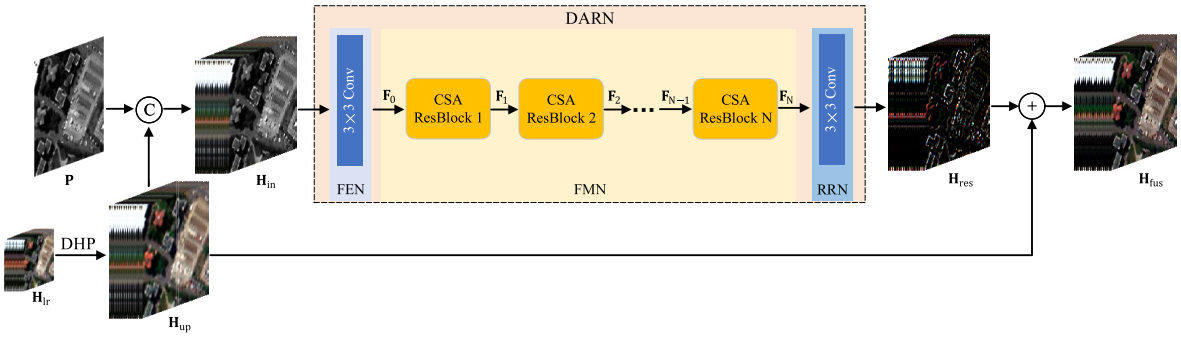



Fig. 1. Overall flowchart of the proposed DHP-DARN method for HS pansharpening. © and ⊕ denote concatenation operation and elementwise sum operation, respectively.

Table I summarizes the above CNN-based remote sensing image fusion methods. Nevertheless, most of the previous CNN models may not sufficiently exploit the intrinsic characteristics of remotely sensed images (e.g., the high correlation among spectral bands and the nonlocal similarity over spatial positions). These approaches lack the ability to make a distinction between useful and trivial information of extracted features and deal with all the components equally, thus hindering the fusion performance. Recently, the attention mechanism has been applied to CNNs, which allows the network to selectively focus on the most informative features. For the HS pansharpening task, there is distinct information of inter- and intrafeature maps, which has different contributions to the fusion. The integration of both CA and SA modules into CNN would be promising for the HS pansharpening. With these considerations, we propose the DARN with CSA ResBlocks, which will be detailed in Section III-B.

## III. PROPOSED METHOD

Fig. 1 illustrates the overall flowchart of the proposed DHP-DARN method for HS pansharpening, which mainly consists of three parts: upsampling the LR-HSI via DHP, mapping the residual HSI through DARN, and obtaining the fused HSI by summation operation. First, the LR-HSI $\mathbf{H}_{\text{lr}} \in \mathbb{R}^{m\times n\times b}$ with $m \times n$ pixels and $b$ bands is upsampled to the scale of the HR-PAN image $\mathbf{P} \in \mathbb{R}^{\beta m\times \beta n\times 1}$ by using the DHP algorithm ($\beta$ denotes the ratio between the spatial resolution of $\mathbf{P}$ and $\mathbf{H}_{\text{lr}}$), which can provide the better quality of the upsampled HSI $\mathbf{H}_{\text{up}} \in \mathbb{R}^{\beta m\times \beta n\times b}$. Then, $\mathbf{H}_{\text{up}}$ is concatenated with $\mathbf{P}$ along the spectral dimension, thus forming the input (denoted by $\mathbf{H}_{\text{in}} \in \mathbb{R}^{\beta m\times \beta n\times (b+1)}$) of the DARN that includes three subnetworks: a feature extraction network (FEN), a feature mapping network (FMN), and a residual reconstruction network (RRN). The FEN applies one convolutional (Conv) layer with a kernel size of $3 \times 3$ to extract the initial feature $\mathbf{F}_0$. To map more informative features for the fusion process, $\mathbf{F}_0$ is fed into the succeeding FMN containing $N$ stacked CSA ResBlocks, which can adaptively highlight significant spectral and spatial information. Subsequently, the residual HSI $\mathbf{H}_{\text{res}}$ between $\mathbf{H}_{\text{up}}$ and the reference HSI $\mathbf{H}_{\text{ref}}$ is reconstructed by the RRN, which can be implemented using a single Conv layer with $b$ filters. Finally, the fused HSI $\mathbf{H}_{\text{fus}} \in \mathbb{R}^{\beta m\times \beta n\times b}$ can be easily obtained by the summation of $\mathbf{H}_{\text{up}}$ and the output residual HSI of the DARN. More details about the three main parts of our newly proposed DHP-DARN architecture for HS pansharpening will be given in Sections III-A–III-C.

### A. Upsampling the LR-HSI via DHP

As shown in Fig. 1, $\mathbf{H}_{\text{lr}}$ is first upsampled to the same size with $\mathbf{P}$ via the DHP method, which is different from bicubic interpolation or LapSRN [23] used in recent CNN-based HSI fusion works [30]–[32], [34]. On the one hand, the DHP approach can produce impressive results for upsampling HSIs without the requirement of gathering large data sets for training, which has a high advantage in the domain of HSI, due to the fact that the openly available HS data sets are very limited. In contrast, the upsampling performance of the LapSRN method used in [32] is highly relied upon learning from large RGB image data sets and the capability

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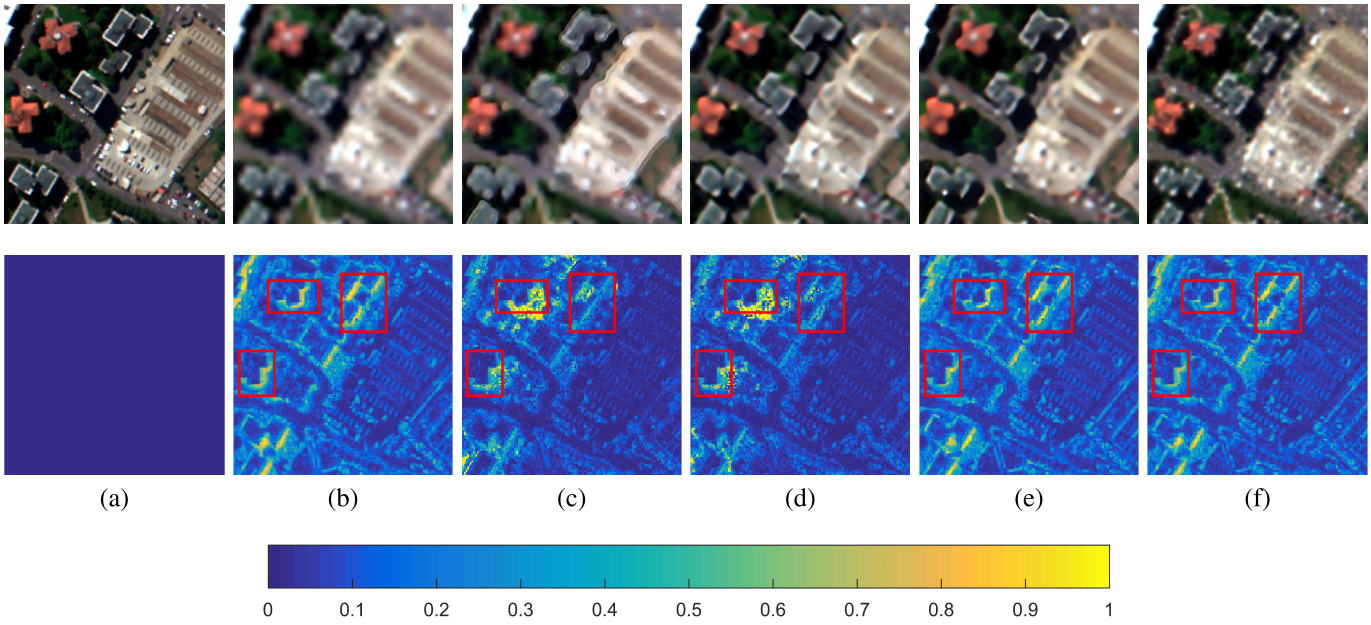



Fig. 2. (First row) Upsampled images of *PaviaCenter_Patch11* (an HSI of the Pavia Center data set which will be introduced in Section IV-A). (Second row) Corresponding spectral angle mappings of different upsampling methods. (a) Reference. (b) Bicubic. (c) LSE. (d) TSE. (e) LapSRN. (f) DHP.

of knowledge adaptation. On the other hand, both the bicubic and LapSRN methods conduct upsampling on each band of the LR-HSI successively, thus ignoring the high spectral correlation of HSIs, which may lead to spectral distortions. Moreover, some spatial details of the LR-HSI are inevitably lost during the process of interpolation. To show the better improvement of spatial details and spectral fidelity of our DHP method, an experimental comparison is conducted on an HSI of the Pavia Center data set, which will be introduced in Section IV-A. Inspired by the experimental discussion of [39], in addition to bicubic and LapSRN methods, two competitive unsupervised SR algorithms (LSE [40] and TSE [41]) are also included for comparison. According to the upsampled results and the corresponding spectral angle mappings displayed in Fig. 2, the DHP method shows finer spatial details and smaller spectral angle mappings than both of the unsupervised and the supervised SR methods do, which validate the superior performance of our DHP method for upsampling the LR-HSI. Note that our DHP method can be fairly compared to these unsupervised SR methods since none of them use other data than the given $\mathbf{H}_{\text{lr}}$ to fulfill the upsampling process.

Fig. 3 illustrates the encoder–decoder U-Net architecture with skip connections that we used in the DHP method, which includes five downsampling operators, five upsampling operators, and five skip-connection operators, corresponding to $d[p]$, $u[p]$, and $s[p]$ $(p = 1, 2, \ldots, 5)$, respectively. The network is conditioned on the degraded image $\mathbf{H}_{\text{lr}}$, taking a fixed tensor $\mathbf{Z}$ filled with uniform noise in the range of 0–0.1 as input, where $\mathbf{Z}$ is of a shape equal to the output $\mathbf{H}_{\text{up}}$. Each operator $d$ consists of a downsampling module with a Lanczos2 kernel [42] followed by batch normalization (BN), LeakyReLU [43] function, and a Conv layer using 128 kernels with a size of $3 \times 3$, which can be regarded as an encoder to extract effective features. For each operator $u$ serving as a decoder to reconstruct the image, there is a bilinear upsampling module and two Conv layers, which have the same number of filters and the same filter size as that of operator $d$. In addition, to alleviate the problem of gradient vanishing and improve the reconstruction accuracy, we add five $s$ operators between the downsampling path and the upsampling path, each of which applies a Conv layer containing four kernels with a size of $1 \times 1$ followed by BN and LeakyReLU operations.

TABLE II
AVERAGE QUANTITATIVE RESULTS OF DHP-$L_2$ AND DHP-$L_1$ METHODS ON THE PAVIA CENTER DATA SET

| Method | CC (↑) | SAM (↓) | RMSE (↓) | ERGAS (↓) | PSNR (↑) |
|---|---|---|---|---|---|
| DHP-$L_2$ | 0.8258 | 6.8813 | 0.0482 | 8.2088 | **21.0212** |
| DHP-$L_1$ | **0.8259** | **6.8709** | **0.0481** | **8.1795** | 20.9959 |

Different from using the $L_2$-distance as the energy function in [37] and [38], we consider utilizing the $L_1$ loss function, as suggested in [36], [44], and [45], to minimize the error between the downsampled version of $\mathbf{H}_{\text{up}}$ and input $\mathbf{H}_{\text{lr}}$, which also prevails in our experiments. To illustrate the effectiveness of $L_1$ loss, the average quantitative results of the DHP method trained with $L_2$ loss (denoted by DHP-$L_2$) and the DHP method using $L_1$ loss (termed as DHP-$L_1$) on the testing set from the Pavia Center data set are exhibited in Table II, where five quantitative metrics that will be described in Section IV-B are included to evaluate the SR performance. The best indexes are highlighted in bold. It can be observed from Table II that the upsampled HSIs produced by the DHP-$L_1$ approach obtains better evaluation results in terms of most quantitative metrics, especially in the aspect of spectral preservation. Therefore, we utilize the DHP-$L_1$ method to provide higher quality upsampled HSIs. The Adam algorithm with a learning rate of 0.01 is used for optimization. The number of optimization steps is empirically set to $1 \times 10^4$

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for each HSI since no notable decay of loss is observed when employing more iterations. Finally, the DHP method can generate the desired $\mathbf{H}_{\text{up}}$ with satisfactory performance, which is independent of learning from training sets.

### B. Mapping the Residual HSI Through DARN

In this section, we describe our DARN from the whole to the part. We first outline the architecture of our DARN and then present the details of two attention modules, i.e., CA module and SA module. Finally, we give the description of the CSA ResBlock that integrates CA and SA modules into the ResBlock to compute complementary attentions concurrently, which can selectively highlight the most relevant information for boosting the performance of HS pansharpening.

*1) Network Architecture:* The proposed DARN consists of a FEN, an FMN, and an RRN, as illustrated in Fig. 1. The input of the DARN, denoted by $\mathbf{H}_{\text{in}}$, is obtained by concatenating the generated $\mathbf{H}_{\text{up}}$ with $\mathbf{P}$ along the spectral axis. The DARN starts with a FEN, where one Conv layer is applied to extract the initial feature representation

$$\mathbf{F}_0 = f_{\text{FEN}}(\mathbf{H}_{\text{in}}) \tag{4}$$

where $f_{\text{FEN}}(\cdot)$ indicates the convolution operation of the FEN and $\mathbf{F}_0$ denotes the extracted feature transformed from $\mathbf{H}_{\text{in}}$. $\mathbf{F}_0$ is then fed into the subsequent FMN, where $N$ CSA ResBlocks are stacked to enhance the representational power of the network. This can be formulated as follows:

$$\mathbf{F}_N = f_{\text{FMN}}(\mathbf{F}_0) = f_{R,N}(f_{R,N-1}(\ldots f_{R,1}(\mathbf{F}_0)\ldots)) \tag{5}$$

where $f_{\text{FMN}}(\cdot)$ represents the feature mapping subnetwork. $f_{R,N}(\cdot)$ is the operation of $N$th CSA ResBlock, which combines two types of attention modules, i.e., CA module and SA module, into the ResBlock to recalibrate the previous feature maps adaptively. After flowing through all the $N$ CSA ResBlocks, the highly informative feature $\mathbf{F}_N$ is generated. In the following RRN, a Conv layer with $b$ filters is employed to recover the spectral dimension and reconstruct the residual HSI:

$$\hat{\mathbf{H}}_{\text{res}} = f_{\text{RRN}}(\mathbf{F}_N) = f_{\text{DARN}}(\mathbf{H}_{\text{in}}) \tag{6}$$

where $f_{\text{RRN}}(\cdot)$ and $f_{\text{DARN}}(\cdot)$ denote the function of the subnetwork (RRN) and the whole network (DARN), respectively. $\hat{\mathbf{H}}_{\text{res}}$ is the reconstructed residual HSI of the DARN, which should match $\mathbf{H}_{\text{res}}$ as closely as possible.

To this end, we utilize the $L_1$ loss function to optimize our DARN, which has been demonstrated as a superior choice for remote sensing image SR [36], [45] and also experimentally verified to be effective for improving the fusion accuracy in this article. For a training set $\{\mathbf{H}_{\text{in}}^k, \mathbf{H}_{\text{res}}^k\}_{k=1}^l$ where $\mathbf{H}_{\text{in}}^k$ is the $k$th input image, $\mathbf{H}_{\text{res}}^k$ is the corresponding residual HSI between $\mathbf{H}_{\text{up}}$ and $\mathbf{H}_{\text{ref}}$, and $l$ is the number of training pairs. The $L_1$ loss function is defined as follows:

$$L(\Theta) = \frac{1}{l}\sum_{k=1}^{l}\left\| f_{\text{DARN}}\left(\mathbf{H}_{\text{in}}^k\right) - \mathbf{H}_{\text{res}}^k\right\|_1 \tag{7}$$

where $\Theta$ represents the parameters of our DARN.

*2) CA Module:* The features extracted by the FEN contain distinct information across channels with different contributions to the fusion process. Thus, more attention should be paid to the informative feature maps, which are highly related to the spectral components of the input data, while the redundant ones should be suppressed. This can be implemented by utilizing the interdependencies between channels. Hence, we construct a CA module to exploit the interchannel relationships for HS pansharpening.

The structure of the CA module is depicted in Fig. 4. Let $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_C]$ denote the input tensor of the CA module, which consists of $C$ feature maps with a spatial size of $H \times W$. To compute the CA and aggregate the spatial information efficiently, a global average pooling layer is first adopted, thus producing the vector $\mathbf{v} \in \mathbb{R}^C$ with its $c$th element

$$v_c = \frac{1}{H \times W}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbf{u}_c(i, j) \tag{8}$$

where $\mathbf{u}_c(i, j)$ is the value of position $(i, j)$ in the $c$th channel $\mathbf{u}_c$, and thus, the global spatial information is squeezed into the statistic $\mathbf{v}$. To make full use of the aggregated information and generate the CA mask $\mathbf{M}_{\text{CA}}$, a gating mechanism with the sigmoid activation $\sigma(\cdot)$ is employed to $\mathbf{v}$

$$\mathbf{M}_{\text{CA}} = \sigma(\Omega_{\text{CA},2}(\delta(\Omega_{\text{CA},1}\mathbf{v}))) \tag{9}$$

where $\delta(\cdot)$ represents the ReLU function, $\Omega_{\text{CA},1} \in \mathbb{R}^{(C/r)\times C}$ and $\Omega_{\text{CA},2} \in \mathbb{R}^{C\times(C/r)}$ are the weights of the first and the second Conv layers in the CA module (both layers comprise kernels of size $1 \times 1$), followed by $\delta(\cdot)$ and $\sigma(\cdot)$ operators, respectively. Bias terms are omitted for simplicity. To reduce the computational cost and improve the nonlinearity of the model, the number of channels is first decreased by the first Conv layer with a reduction ratio $r$ and then increased back to $C$ by the following Conv layer after being activated by the ReLU operation. By obtaining the CA map $\mathbf{M}_{\text{CA}} \in \mathbb{R}^C$ with values varying between 0 and 1, the resultant output of the CA module can be acquired as follows:

$$\begin{aligned}\mathbf{U}_{\text{CA}} &= f_{\text{CA}}(\mathbf{M}_{\text{CA}}, \mathbf{U}) \\ &= [\mathbf{M}_{\text{CA},1}\mathbf{u}_1, \mathbf{M}_{\text{CA},2}\mathbf{u}_2, \ldots, \mathbf{M}_{\text{CA},C}\mathbf{u}_C]\end{aligned} \tag{10}$$

where $f_{\text{CA}}(\cdot)$ represents the multiplication function for the input $\mathbf{U}$ and the corresponding CA map $\mathbf{M}_{\text{CA}}$, and $\mathbf{U}_{\text{CA}}$ is the channelwise refined output.

With the proposed CA module, the network is able to selectively suppress the less important channels and emphasize the more relevant ones, which is beneficial to the improvement of discriminability for spectral features.

*3) SA Module:* Different from the CA that squeezes the spatial information into a channel descriptor, the SA aims at highlighting the most informative regions of the input since each feature map contains diverse information over spatial locations. For instance, areas of edges and textures are expected to incorporate high-frequency information, which is usually more difficult to sharpen, while the smoother regions are with abundant low-frequency components. Therefore, we build an SA module to fully utilize the interspatial relationships, which is complementary to the CA module.

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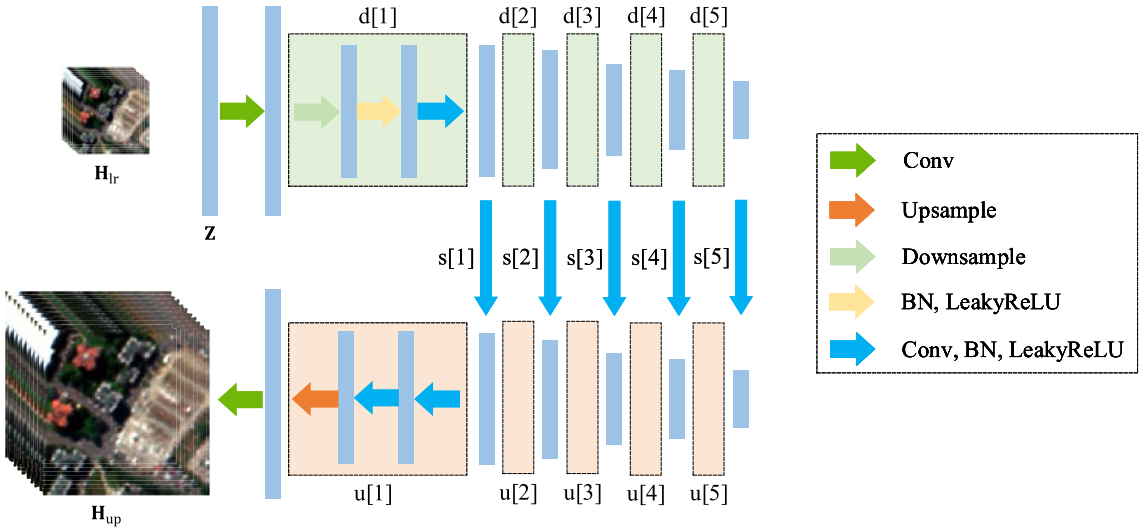



Fig. 3. U-Net architecture used in the DHP method.

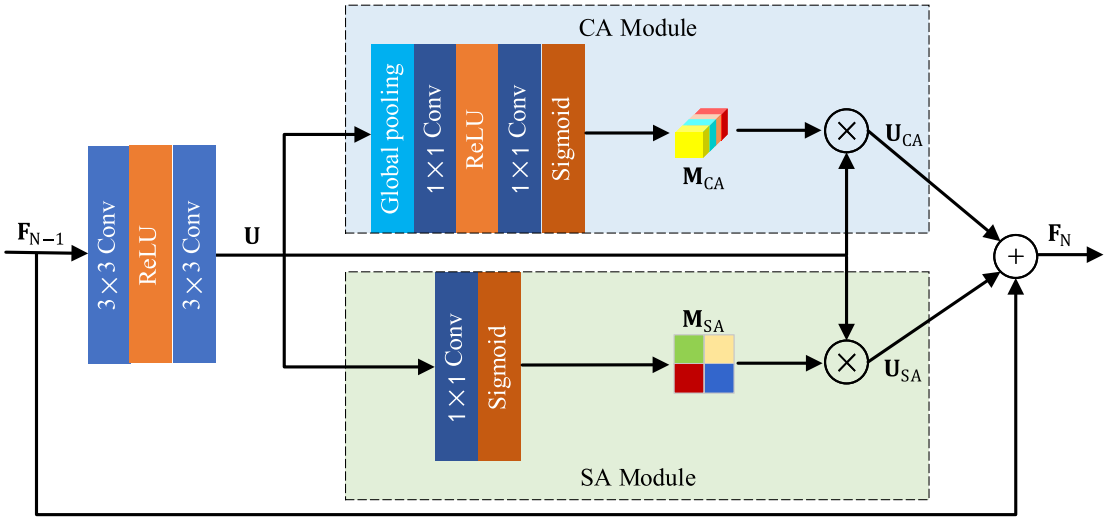



Fig. 4. Diagram of the CSA ResBlock, which integrates the CA module and the SA module into the ResBlock. ⊗ and ⊕ denote the elementwise multiplication and the elementwise sum operation, respectively.

For the SA module as illustrated in Fig. 4, we consider an alternative representation of the input tensor $\mathbf{U} = [\mathbf{u}^{1,1}, \mathbf{u}^{1,2}, \ldots, \mathbf{u}^{i,j}, \ldots, \mathbf{u}^{H,W}]$, where $\mathbf{u}^{i,j} \in \mathbb{R}^C$ denotes the feature vector at the spatial location $(i, j)$. To yield the SA map $\mathbf{M}_{\mathrm{SA}}$, a simple $1 \times 1$ Conv layer with weight $\Omega_{\mathrm{SA}} \in \mathbb{R}^{1\times1\times C}$ is performed on $\mathbf{U}$, followed by the sigmoid function $\sigma(\cdot)$ to rescale the attentive map to the range of $[0, 1]$, which can be formulated as follows:

$$\mathbf{M}_{\mathrm{SA}} = \sigma(\Omega_{\mathrm{SA}}\mathbf{U}) \tag{11}$$

where $\mathbf{M}_{\mathrm{SA}} \in \mathbb{R}^{H\times W}$ is the SA mask, which is then used to recalibrate $\mathbf{U}$ spatially

$$\begin{aligned}\mathbf{U}_{\mathrm{SA}} &= f_{\mathrm{SA}}(\mathbf{M}_{\mathrm{SA}}, \mathbf{U}) \\ &= \left[\mathbf{M}_{\mathrm{SA}}^{1,1}\mathbf{u}^{1,1}, \mathbf{M}_{\mathrm{SA}}^{1,2}\mathbf{u}^{1,2}, \ldots, \mathbf{M}_{\mathrm{SA}}^{H,W}\mathbf{u}^{H,W}\right]\end{aligned} \tag{12}$$

where $f_{\mathrm{SA}}(\cdot)$ refers to the elementwise multiplication for the input $\mathbf{U}$ and the obtained SA map, and $\mathbf{U}_{\mathrm{SA}}$ is the recalibration result of the SA module. Each $\mathbf{M}_{\mathrm{SA}}^{i,j}$ corresponds to the SA weight for the spatial location $(i, j)$ of $\mathbf{U}$.

With the process described above, the SA module can adaptively provide more importance to the relevant regions that are closely related to spatial details enhancement for HS pansharpening, which could be combined with the CA module to further boost the representational ability of our DARN.

*4) CSA ResBlock:* In order to take full advantage of the two kinds of attention mechanisms, we integrate the CA module and the SA module into the ResBlock [22], thus adapting it to the CSA ResBlock for HS pansharpening, as shown in Fig. 4.

We denote the input of the $N$th CSA ResBlock as $\mathbf{F}_{N-1}$, which is first passed through two $3 \times 3$ Conv layers that are connected via a ReLU activation for deep feature extraction. In this way, a set of more refined features denoted by $\mathbf{U}$ is produced, which is then concurrently fed into the CA module and the SA module to be recalibrated. By means of the operations

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of two types of attention modules as described in Sections III-B2 and III-B3, two sets of attention-weighted feature maps termed as $\mathbf{U}_{\mathrm{CA}}$ and $\mathbf{U}_{\mathrm{SA}}$, respectively, are generated. Moreover, we adopt the skip connection in each CSA ResBlock, which effectively combines low- and high-level features and further improves the performance and efficiency of the DARN. At last, the output of the $N$th CSA ResBlock is generated through the aggregation of the input $\mathbf{F}_{N-1}$ and the summation of the two attention modules

$$\mathbf{F}_N = \mathbf{F}_{N-1} + (\mathbf{U}_{\mathrm{CA}} + \mathbf{U}_{\mathrm{SA}}) \tag{13}$$

where $\mathbf{F}_{\mathrm{N}}$ represents the output of the $N$th CSA ResBlock. By incorporating two complementary attention modules into the ResBlock, i.e., CA and SA modules, the CSA ResBlock can not only encourage the network to adaptively focus on more useful features along both channel and spatial dimension simultaneously but also exploit the benefits of the skip connection to facilitate the training of deep CNNs. In addition, both of our attention modules are designed to be lightweight, which can be conveniently employed to multiple CSA ResBlocks to capture more meaningful information for HS pansharpening.

### C. Obtaining the Fused HSI by Summation Operation

After carrying out the first two parts of the DHP-DARN framework, the DHP upsampling result $\mathbf{H}_{\mathrm{up}}$ and the output residual HSI $\hat{\mathbf{H}}_{\mathrm{res}}$ of the DARN are created. Finally, we can obtain the fused HSI $\mathbf{H}_{\mathrm{fus}}$ by the summation operation of $\mathbf{H}_{\mathrm{up}}$ and $\hat{\mathbf{H}}_{\mathrm{res}}$ (as outlined in Fig. 1)

$$\mathbf{H}_{\mathrm{fus}} = \mathbf{H}_{\mathrm{up}} + \hat{\mathbf{H}}_{\mathrm{res}}. \tag{14}$$

## IV. Experimental Results and Discussion

In this section, we first present the experimental data sets and the quantitative metrics used for experiments and performance evaluation. Then, we provide the description of the experimental settings. Subsequently, we compare our DHP-DARN method with several state-of-the-art HS pansharpening algorithms on both simulated and real HS data sets. Finally, we show the results of the ablation experiments to demonstrate the effectiveness of the CSA ResBlock and the residual learning.

### A. Experimental Data Sets

To evaluate the performance of our proposed DHP-DARN method for HS pansharpening, a series of experiments was conducted on three simulated HS data sets and one real HS data set, which are described in detail in the following.

1) *CAVE Data Set:* The CAVE [46] data set contains 32 HSIs of real-world objects acquired in the laboratory environment. All the HSIs in this data set have 31 spectral bands ranging from 400 to 700 nm at a wavelength interval of 10 nm, each band of which has a spatial dimension of $512 \times 512$ pixels. With the available HSIs in the CAVE data set serving as the reference images, the simulated experimental LR-HSIs of size $128 \times 128$ and HR-PAN images with $512 \times 512$ pixels can be generated based on the Wald's protocol [47]. The training set of the CAVE data set contains 22 randomly selected HSIs (about 70%), and the remaining ten images are utilized for testing.
2) *Pavia Center Data Set:* The image of Pavia Center with an original size of $1096 \times 1096 \times 115$ was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over the city of Pavia, Italy [48]. Thirteen noisy bands were removed, and the remaining 102 spectral channels covering a spectral range from 430 to 860 nm were processed. In addition, a 381-pixel-wide poor quality area of the image was discarded, thus resulting in a "two-part" image with $1096 \times 715$ spatial pixels. For the experimental design, we used only the top-left corner of size $960 \times 640 \times 102$ of the image and then partitioned the selected part into 24 cubic-patches of size $160 \times 160 \times 102$ with no overlap, which constituted the Pavia Center data set. In this data set, the dimensions of the test PAN images are $160 \times 160$ pixels with a spatial resolution of 1.3 m, and the HSIs used for experiments are of size $40 \times 40$ with a spatial resolution of 5.2 m. We randomly choose 17 cubic-patches (about 70%) for training, and the rest seven HSIs comprise the testing set.
3) *Botswana Data Set:* The Botswana image was collected via the Hyperion sensor on the Earth Observing One (EO-1) satellite at a 30-m spatial resolution over the Okavango Delta, Botswana [49]. This image originally had $1496 \times 256$ pixels with 242 bands covering the spectral range of 400–2500 nm in 10-nm windows. After removing the uncalibrated and noisy bands, the remaining 145 bands were used for the study. For convenience, only the top-left part of the image with a spatial coverage of $1200 \times 240$ was utilized for our experiments, which was then divided into 20 cubic-patches of size $120 \times 120 \times 145$ without overlapping, thus forming the Botswana data set. For the test HSIs and PAN images of the Botswana data set, the spatial dimensions are $40 \times 40$ and $120 \times 120$, respectively. For the Botswana data set, 14 cubic-patches (70%) are selected at random for training, and the remaining six HSIs are used for testing.
4) *Los Angeles Data Set:* The Los Angeles data set taken over a port in the city of Los Angeles consisted of an LR-HSI and the corresponding HR-PAN image, which was obtained by two instruments on the EO-1 satellite, i.e., the Hyperion and the advanced land imager (ALI) [50]. Specifically, the HSI including 242 bands was provided by the Hyperion with a spatial resolution of 30 m, and the PAN image was acquired by the ALI with a spatial resolution of 10 m. According to the candidate features of the Botswana data set, uncalibrated and noisy spectral bands of the Los Angeles data set were removed, thus leaving 145 bands for experimentation. Thus, the well-trained model on the Botswana data set can be generalized to the pansharpening case of this Los Angeles data set, which is a real HS data set. The size of the test HSI is $120 \times 120 \times 145$, and the dimension of the experimental PAN image is $360 \times 360$.

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The CAVE, Pavia Center, and Botswana data sets are all semisynthetic data sets. Given the reference HR-HSIs from the three data sets, the HR-PAN images and LR-HSIs can be simulated. The PAN image was obtained by averaging the visible bands of the reference HR-HSI. To create an LR-HSI, we spatially blurred the reference one with an $8 \times 8$ Gaussian filter and then downsampled the result. The scaling factor was set to 4 for the CAVE and the Pavia Center data set. For the Botswana data set, the reference images were downsampled by a factor of 3. The Los Angeles data set is a real data set to evaluate the fusion performance of the proposed method on the actual HSI.

It is worth noting that our experiments are conducted on both simulated and real HS data sets with different ratios between the spatial resolution of the PAN image and the HSI, which can fully demonstrate the generalization capability of the proposed DHP-DARN method.

### *B. Quantitative Metrics*

In order to perform the objective fusion quality assessment of the simulated data sets, the fusion results generated by fusing the simulated HR-PAN images and LR-HSIs are compared with the reference ones. Let $X_{h,w}$ denote the pixel value of the reference image $\mathbf{H}_{\text{ref}} \in \mathbb{R}^{\beta m \times \beta n \times b}$, and $\hat{X}_{h,w}$ represent the pixel value of the fused image $\mathbf{H}_{\text{fus}} \in \mathbb{R}^{\beta m \times \beta n \times b}$, where $h = 1, 2, \ldots, \beta m$ and $w = 1, 2, \ldots, \beta n$, respectively. The definitions of $\beta$, $m$, $n$, and $b$ remain the same as in Section III. Five complementary and widely used quantitative full-reference metrics are utilized to assess the fusion performance, which are detailed as follows.

1) *Cross Correlation (CC) [51]:* CC that gives the measurement of spatial distortion is defined as follows:

$$\text{CC} = \frac{\sum_{h,w}[(X_{h,w} - \mu)(\hat{X}_{h,w} - \hat{\mu})]}{\sqrt{\sum_{h,w}[(X_{h,w} - \mu)^2] \sum_{h,w}[(\hat{X}_{h,w} - \hat{\mu})^2]}} \tag{15}$$

where $\mu$ and $\hat{\mu}$ are the mean values of $\mathbf{H}_{\text{ref}}$ and $\mathbf{H}_{\text{fus}}$, respectively. A CC value of 1 indicates that the two HSIs are highly correlated.

2) *Spectral Angle Mapper (SAM) [51]:* SAM is defined as the angle between the reference and the fused spectral vectors

$$\text{SAM} = \arccos\left(\frac{\langle \mathbf{X}_{h,w}, \hat{\mathbf{X}}_{h,w} \rangle}{\|\mathbf{X}_{h,w}\|_2 \|\hat{\mathbf{X}}_{h,w}\|_2}\right) \tag{16}$$

in which $\langle \cdot, \cdot \rangle$ denotes the inner product, $\| \cdot \|_2$ is the $L_2$-norm, and $\mathbf{X}_{h,w}$ and $\hat{\mathbf{X}}_{h,w}$ are the vector-based representations of $X_{h,w}$ and $\hat{X}_{h,w}$, respectively. SAM is usually averaged over the whole spatial domain to reflect the spectral distortion. A smaller SAM value means less spectral distortion.

3) *Root-Mean-Squared Error (RMSE) [51]:* RMSE is defined to calculate the $L_2$ difference between the reference and the pan-sharpened images

$$\text{RMSE} = \sqrt{\frac{1}{\beta m \beta n} \sum_{h=1}^{\beta m} \sum_{w=1}^{\beta n} (X_{h,w} - \hat{X}_{h,w})^2}. \tag{17}$$

A smaller RMSE value shows that the fused image is closer to the reference one.

4) *Erreur Relative Globale Adimensionnelle De Synthèse (ERGAS) [51]:* ERGAS offers a global quality evaluation of the fused image, which is formulated as follows:

$$\text{ERGAS} = 100\beta \sqrt{\frac{1}{b} \sum_{q=1}^{b} \left[\frac{\text{RMSE}_q}{\mu\left(\mathbf{H}_{\text{ref}}^q\right)}\right]^2} \tag{18}$$

where $\text{RMSE}_q$ and $\mu(\mathbf{H}_{\text{ref}}^q)$ are the RMSE and mean values of the $q$th band, respectively. The optimal value of ERGAS is 0.

5) *Peak SNR (PSNR) [17]*: PSNR is another index to assess the fusion quality of each band, and the average PSNR of all bands is given by

$$\text{PSNR} = \frac{1}{b} \sum_{q=1}^{b} \left[10 \cdot \log_{10}\left(\frac{\max\left(\mathbf{H}_{\text{ref}}^q\right)}{\text{RMSE}_q}\right)^2\right] \tag{19}$$

where $\max(\mathbf{H}_{\text{ref}}^q)$ is the maximum pixel value in the $q$th band of $\mathbf{H}_{\text{ref}}$. A larger value of PSNR represents a higher reconstruction quality in spatial information of the fusion result.

In addition, we also investigate the CC and PSNR plots over all the spectral bands to perform the band-dependent quality evaluation of the fused image.

For the real data set of Los Angeles, we adopt the *quality with no reference* (QNR) [52] and two derived metrics, i.e., $D_\lambda$ and $D_S$, for quantitative evaluation since the reference HR-HSI is not available. The QNR metric is defined as follows:

$$\text{QNR} = (1 - D_\lambda)^\eta (1 - D_S)^\rho \tag{20}$$

where $\eta$ and $\rho$ are two coefficients, and $D_\lambda$ and $D_S$ are defined to quantify the spectral and spatial distortion, respectively. Specifically, the spectral distortion is calculated by

$$D_\lambda = \sqrt[\varepsilon]{\frac{1}{b(b-1)} \sum_{q_1=1}^{b} \sum_{q_2=1, q_2 \neq q_1}^{b} \left|\text{Q}_{\text{lr}}^{q_1,q_2} - \text{Q}_{\text{fus}}^{q_1,q_2}\right|^\varepsilon} \tag{21}$$

where the parameter $\varepsilon$ is usually set to 1, and $\text{Q}_{\text{lr}}^{q_1,q_2} = \text{Q}(\mathbf{H}_{\text{lr}}^{q_1}, \mathbf{H}_{\text{lr}}^{q_2})$ and $\text{Q}_{\text{fus}}^{q_1,q_2} = \text{Q}(\mathbf{H}_{\text{fus}}^{q_1}, \mathbf{H}_{\text{fus}}^{q_2})$ is defined to utilize the *Q-index* [53] to calculate the dissimilarities between couples of bands for $\mathbf{H}_{\text{lr}}$ and $\mathbf{H}_{\text{fus}}$, respectively. Thus, the relevance among spectral bands should be preserved during the fusion. Given two images $\mathbf{I}$ and $\mathbf{J}$, the Q-index with an ideal value of 1 is expressed as follows:

$$\text{Q}(\mathbf{I}, \mathbf{J}) = \frac{\varphi_{\mathbf{IJ}}}{\varphi_{\mathbf{I}} \varphi_{\mathbf{J}}} \frac{2\overline{\mathbf{I}}\,\overline{\mathbf{J}}}{(\overline{\mathbf{I}})^2 + (\overline{\mathbf{J}})^2} \frac{2\varphi_{\mathbf{I}} \varphi_{\mathbf{J}}}{\left(\varphi_{\mathbf{I}}^2 + \varphi_{\mathbf{J}}^2\right)} \tag{22}$$

where $\varphi_{\mathbf{IJ}}$ denotes the covariance between $\mathbf{I}$ and $\mathbf{J}$, and $\overline{\mathbf{I}}$ and $\varphi_{\mathbf{I}}$ are the sample mean and variance of $\mathbf{I}$. For the spatial distortion $D_S$, it is calculated in the form

$$D_S = \sqrt[\kappa]{\frac{1}{b} \sum_{q=1}^{b} \left|\text{Q}\left(\mathbf{H}_{\text{fus}}^q, \mathbf{P}\right) - \text{Q}\left(\mathbf{H}_{\text{lr}}^q, \mathbf{P}_{\text{lr}}\right)\right|^\kappa} \tag{23}$$

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where $\kappa$ is typically set as 1, and $\mathbf{P}_{\text{lr}}$ denotes a simulated LR-PAN image with the same scale of the LR-HSI. Accordingly, the QNR index obtains the best value of 1 when both $D_\lambda$ and $D_S$ are 0.

### C. Experimental Setup

The implementation details of the DHP algorithm used for upsampling the LR-HSI have been specified in Section III-A. Here, we mainly give the description of the key parameters settings of our proposed DARN, including model hyperparameters, training settings, and network depth.

Following [35], all Conv layers of the FEN and the FMN in our DARN have $C = 64$ filters with a kernel size of $3 \times 3$, except that in the CA and SA modules, where the size of kernels is set to $1 \times 1$. In addition, the filter numbers of the channel-downscaling layers in the CA modules and the Conv layers in the SA modules are $C/r = 4$ and 1, respectively, i.e., the reduction ratio $r$ is set as 16. Table III lists the average quantitative results of the proposed method with different reduction ratios, where the best indexes are marked in bold. Apparently, most of the fusion results obtain the best evaluation indexes when $r = 16$. Therefore, 16 is the optimal setting for $r$ in the FMN. For the final Conv layer in the RRN, it comprises $b$ filters with a kernel size of $3 \times 3$, which are equal to the number of bands to recover the spectral dimension. Meanwhile, the zero-padding strategy is used for all these $3 \times 3$ Conv layers to keep the size fixed.

For training, both $\mathbf{H}_{\text{in}}$ and $\mathbf{H}_{\text{res}}$ of each training pair are partitioned into patches of size $32 \times 32$. The batch size is set to 128, and the Adam optimizer with a learning rate of $1 \times 10^{-3}$ is adopted to minimize the $L_1$ loss. We terminate the training process at $1.5 \times 10^3$ epochs where the loss stops remarkable decreasing. For testing, the concatenated images from the testing sets are served as the input of the DARN, and the final fused HSIs are compared to their corresponding reference HSIs for quality evaluation. Training and testing of the proposed DARN are both implemented based on the TensorFlow DL framework by using an NVIDIA GeForce GTX 1080Ti GPU.

The depth of the network is directly determined by the number of CSA ResBlocks $N$, which has an important impact on the fusion performance. To study the effect of different block numbers, we report the variation of average quantitative results and training time cost versus the only changing number of blocks on the three simulated HS data sets in Fig. 5, with the other settings of the network keeping the same. Fig. 5(a)–(e) shows the average quantitative result curves of $\mathbf{H}_{\text{fus}}$ as functions of the number of blocks, i.e., CC, SAM, RMSE, ERGAS, and PSNR value, respectively. In general, the better fusion accuracy can be achieved as $N$ increases, which suggests the superiority of a deeper model. Specifically, the quantitative results of almost all the evaluation metrics for the CAVE data set remain stable with changing block numbers. For the Pavia Center data set and the Botswana data set, most metrics achieve their best values when $N = 4$, and employing more blocks does not always deliver a better performance. As depicted in Fig. 5(f), although some indexes can obtain small improvements when the number of blocks increases from 4 to 5 or 6, the training time of the network grows rapidly, especially for the CAVE data set with much more input training patches. Taking both the fusion performance and the time cost into consideration, the number of CSA ResBlocks $N$ is set to 4 in the following experiments, which is appropriate and effective.

TABLE III
AVERAGE QUANTITATIVE RESULTS OF THE PROPOSED METHOD WITH DIFFERENT REDUCTION RATIOS $r$

| Dataset | $r$ | CC (↑) | SAM (↓) | RMSE (↓) | ERGAS (↓) | PSNR (↑) |
|---|---|---|---|---|---|---|
| CAVE | 2 | 0.9988 | 2.7448 | 0.0068 | 1.4121 | 43.9363 |
| | 4 | 0.9987 | 2.7604 | 0.0068 | 1.4041 | 43.9241 |
| | 8 | **0.9989** | 2.7399 | 0.0066 | 1.4064 | 43.9559 |
| | 16 | **0.9989** | **2.7256** | **0.0065** | **1.4026** | **43.9873** |
| | 32 | 0.9988 | 2.7456 | 0.0067 | 1.4131 | 43.9466 |
| Pavia Center | 2 | 0.9795 | 4.1754 | 0.0167 | 2.6369 | 37.1980 |
| | 4 | 0.9802 | 4.1081 | 0.0165 | 2.5833 | 37.5753 |
| | 8 | 0.9803 | 4.1025 | 0.0164 | **2.5701** | 37.5815 |
| | 16 | **0.9804** | **4.0909** | **0.0161** | 2.5771 | **37.6089** |
| | 32 | 0.9779 | 4.4469 | 0.0173 | 2.7800 | 36.6405 |
| Botswana | 2 | 0.9589 | 1.7668 | 0.0109 | 2.5670 | 30.6379 |
| | 4 | 0.9558 | 1.8187 | 0.0114 | 2.7481 | 30.3116 |
| | 8 | 0.9540 | 1.8174 | 0.0113 | 2.7686 | 30.2815 |
| | 16 | **0.9657** | **1.6933** | **0.0106** | **2.2352** | **31.2196** |
| | 32 | 0.9589 | 1.7942 | 0.0112 | 2.5392 | 30.5763 |

### D. Experiments on Three Simulated Hyperspectral Data Sets

In this section, the proposed DHP-DARN method is compared with eight representative classical HS pansharpening approaches belonging to different branches, i.e., GSA [9], PCA [10], SFIM [11], MGH [12], GFPCA [14], BNA [15], BSP [16], and CNMF [17], and two CNN-based state-of-the-art HS pansharpening methods, including DDLPS [32] and DRCNN [34]. Here, experiments are performed on the three simulated HS data sets, i.e., CAVE, Pavia Center, and Botswana data sets, as described in Section IV-A.

*1) CAVE Data Set:* The average quantitative results of different HS pansharpening methods on the testing set from the CAVE data set are shown in Table IV. The best indexes are marked in bold. From Table IV, it can be seen that the proposed DHP-DARN method obtains the biggest CC and PSNR indexes and the smallest SAM, RMSE, and ERGAS values, which indicates that the proposed method achieves the best fusion performance among all the competing methods.

For visual comparison, the image *real_and_fake_peppers* from the testing set of the CAVE data set is selected for presentation, and the reference HR-HSI of *real_and_fake_peppers* is displayed in Fig. 6(a). Fig. 6(b)–(l) shows the visual results obtained by different HS pansharpening methods for the image *real_and_fake_peppers*. One meaningful region of each reconstructed image is marked in a red square for easy comparison. It is observed that the GSA and PCA approaches suffer from severe spectral distortion, such as the body of the peppers. The SFIM and MGH methods cause

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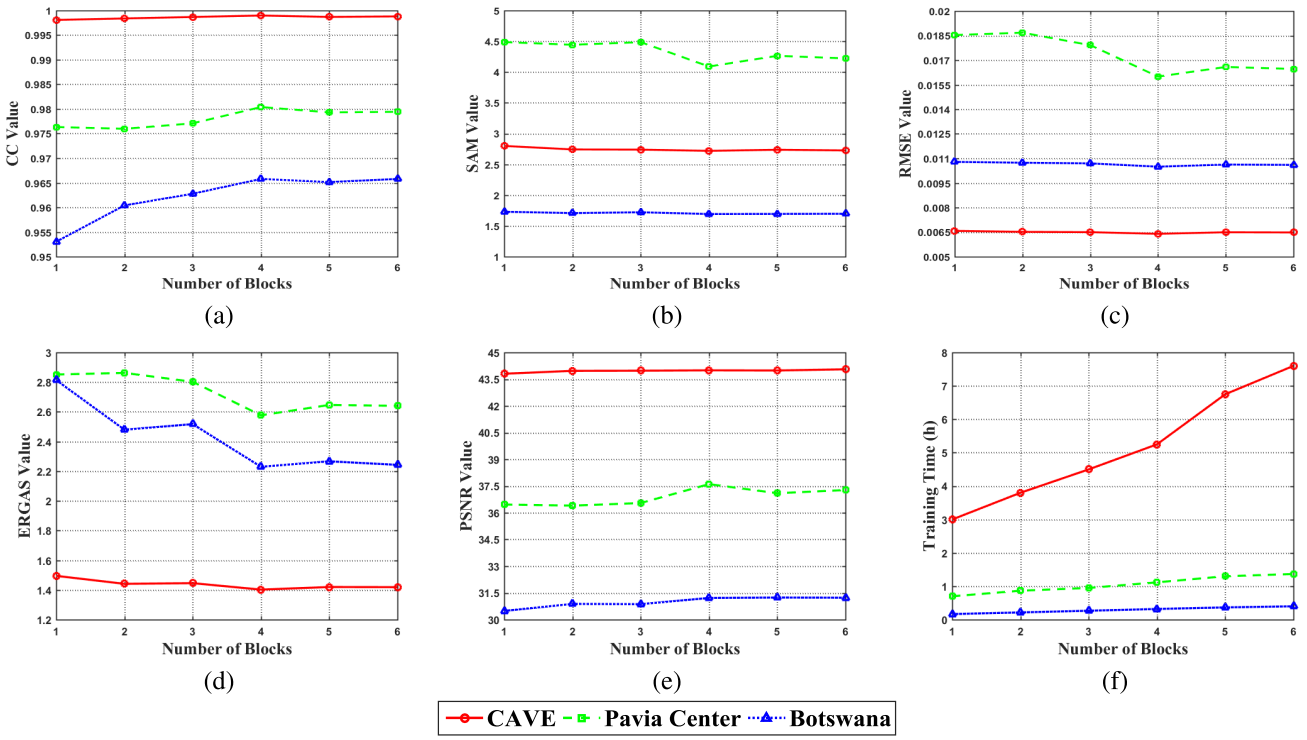



Fig. 5. Average quantitative results of $\mathbf{H}_{\text{fus}}$ as functions of the number of CSA ResBlocks. (a) CC value. (b) SAM value. (c) RMSE value. (d) ERGAS value. (e) PSNR value. (f) Training time ($h$).

TABLE IV
AVERAGE QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE CAVE DATA SET

| Method | CC (↑) | SAM (↓) | RMSE (↓) | ERGAS (↓) | PSNR (↑) |
|---|---|---|---|---|---|
| GSA | 0.9956 | 4.6772 | 0.0134 | 3.1498 | 37.2597 |
| PCA | 0.9891 | 6.5509 | 0.0232 | 4.2222 | 35.0643 |
| SFIM | 0.9881 | 6.2074 | 0.0212 | 5.4605 | 34.6478 |
| MGH | 0.9868 | 6.7060 | 0.0220 | 5.7868 | 34.7478 |
| GFPCA | 0.9908 | 4.1329 | 0.0203 | 4.5890 | 32.3137 |
| BNA | 0.9880 | 3.5921 | 0.0221 | 5.2432 | 31.6036 |
| BSP | 0.9932 | 5.5709 | 0.0159 | 3.9425 | 34.3861 |
| CNMF | 0.9946 | 6.0153 | 0.0144 | 3.1774 | 35.5895 |
| DDLPS | 0.9943 | 4.2529 | 0.0179 | 3.7275 | 36.1835 |
| DRCNN | 0.9949 | 3.7324 | 0.0147 | 3.0306 | 37.6511 |
| Proposed | **0.9989** | **2.7256** | **0.0065** | **1.4026** | **43.9873** |

TABLE V
AVERAGE QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE PAVIA CENTER DATA SET

| Method | CC (↑) | SAM (↓) | RMSE (↓) | ERGAS (↓) | PSNR (↑) |
|---|---|---|---|---|---|
| GSA | 0.9458 | 6.8414 | 0.0298 | 4.4704 | 30.5368 |
| PCA | 0.8874 | 7.6574 | 0.0426 | 6.8316 | 25.6715 |
| SFIM | 0.9434 | 6.2119 | 0.0304 | 4.6274 | 30.0513 |
| MGH | 0.9476 | 6.1889 | 0.0295 | 4.3493 | 31.3292 |
| GFPCA | 0.8705 | 8.5679 | 0.0503 | 7.6964 | 20.8409 |
| BNA | 0.8634 | 6.5049 | 0.0433 | 7.4515 | 20.6931 |
| BSP | 0.9026 | 8.5256 | 0.0418 | 6.5590 | 23.5120 |
| CNMF | 0.9296 | 6.3664 | 0.0346 | 5.3238 | 22.9290 |
| DDLPS | 0.9150 | 7.5713 | 0.0396 | 5.3985 | 28.3788 |
| DRCNN | 0.9360 | 6.6114 | 0.0324 | 4.8672 | 29.6026 |
| Proposed | **0.9804** | **4.0909** | **0.0161** | **2.5771** | **37.6089** |

the loss of spatial details in some regions, especially in the edges and textured areas. The fused results generated by the GFPCA and Bayesian approaches are fuzzy, where the characters are blurred. By contrast, the fused HSIs produced by the CNMF and three CNN-based methods achieve superior performance in spatial details reconstruction and spectral information preservation. As depicted in Fig. 6(l), our proposed method accurately reconstructs the edge details of the marked region.

To further show the effectiveness of the proposed method clearly, the corresponding mean absolute error images between the reconstructed HSIs and the reference image *real_and_fake_peppers* are shown in Fig. 7, where dark blue represents smaller differences. It can be obviously seen that the proposed method achieves the smallest value difference among all the competing methods, which means that the fusion result obtained by the proposed method is the closest to the reference HSI. The abovementioned quantitative results and visual comparison demonstrate that the proposed DHP-DARN method performs well in both objective and subjective evaluations.

*2) Pavia Center Data Set:* Table V lists the average quantitative results of different fusion methods on the testing set from the Pavia Center data set. From Table V, it can be apparently observed that the proposed method achieves the best fusion accuracy with respect to all the

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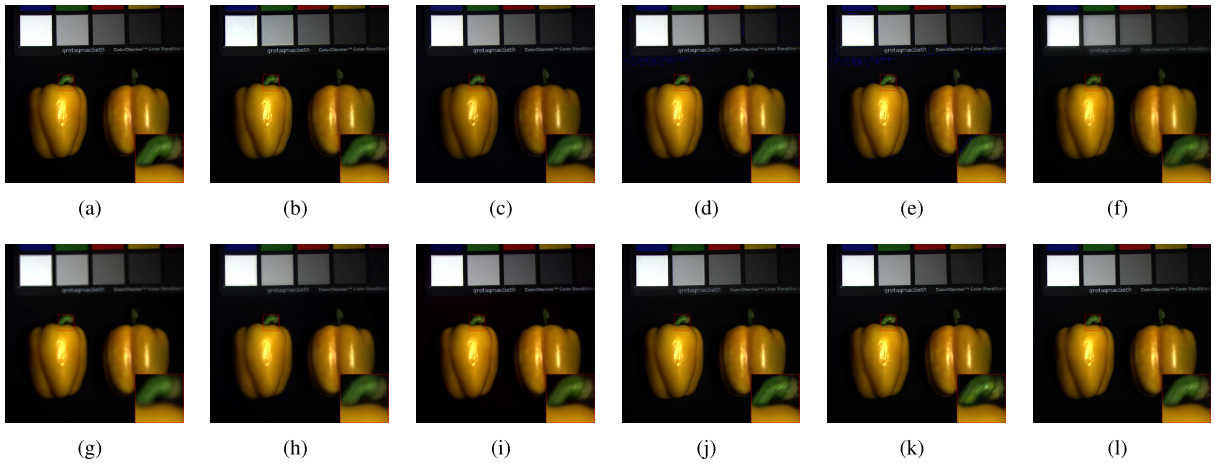



Fig. 6. Visual results obtained by different methods for the *real_and_fake_peppers* (an HSI of the CAVE data set). (a) Reference. (b) GSA. (c) PCA. (d) SFIM. (e) MGH. (f) GFPCA. (g) BNA. (h) BSP. (i) CNMF. (j) DDLPS. (k) DRCNN. (l) Proposed.

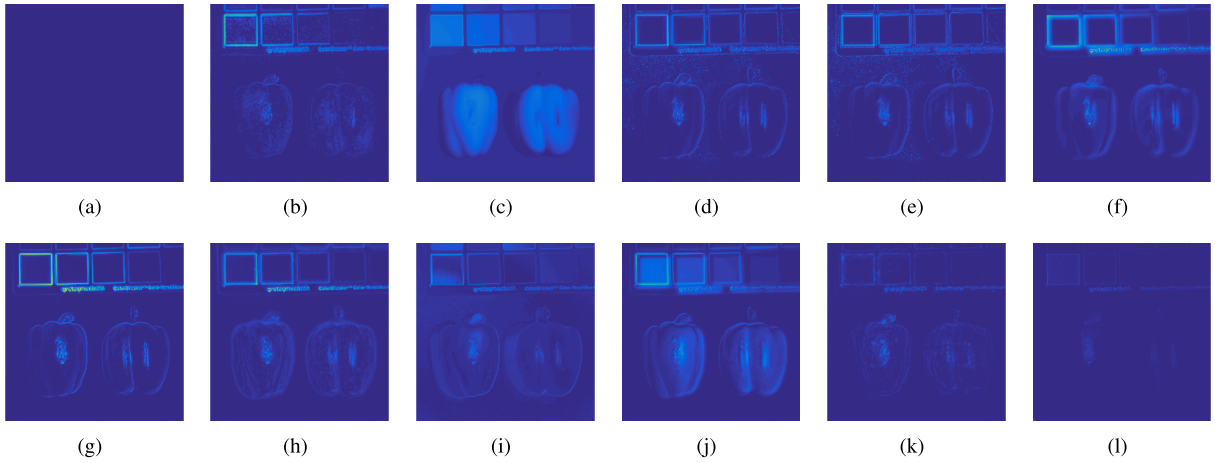



Fig. 7. Mean absolute error images of the competing methods for the *real_and_fake_peppers* (an HSI of the CAVE data set). (a) Reference. (b) GSA. (c) PCA. (d) SFIM. (e) MGH. (f) GFPCA. (g) BNA. (h) BSP. (i) CNMF. (j) DDLPS. (k) DRCNN. (l) Proposed.

TABLE VI
AVERAGE QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE BOTSWANA DATA SET

| Method | CC (↑) | SAM (↓) | RMSE (↓) | ERGAS (↓) | PSNR (↑) |
|---|---|---|---|---|---|
| GSA | 0.9618 | 2.0194 | 0.0140 | 2.2690 | 29.8486 |
| PCA | 0.9623 | 2.2649 | 0.0166 | 2.4786 | 28.3591 |
| SFIM | 0.9444 | 2.1592 | 0.0180 | 2.6321 | 27.5858 |
| MGH | 0.9597 | 2.0554 | 0.0147 | 2.2543 | 29.6406 |
| GFPCA | 0.9096 | 2.8147 | 0.0241 | 3.6132 | 23.8119 |
| BNA | 0.9046 | 2.3983 | 0.0244 | 3.3547 | 24.6955 |
| BSP | 0.8964 | 3.3473 | 0.0246 | 3.7135 | 23.7952 |
| CNMF | 0.9474 | 2.1839 | 0.0155 | 2.6284 | 27.1972 |
| DDLPS | 0.9511 | 2.1156 | 0.0198 | 2.9132 | 26.3369 |
| DRCNN | 0.9164 | 2.3630 | 0.0177 | 3.3377 | 26.7381 |
| Proposed | **0.9657** | **1.6933** | **0.0106** | **2.2352** | **31.2196** |

five quantitative metrics, which indicates that the proposed method can fulfill the HS pansharpening with the least distortion. In addition, the proposed DHP-DARN method greatly outperforms the other two CNN-based HS pansharpening approaches in terms of quantitative evaluations, which further validates the superiority of our proposed attention modules.

Fig. 8(a) shows the reference HR-HSI of the *PaviaCenter_Patch21*, which is the 21st cubic-patch of the testing set from the Pavia Center data set. The visual results generated by different HS pansharpening algorithms for this HSI are displayed in Fig. 8(b)–(l). There is an obvious color difference of the fused images obtained by the CS, hybrid, and CNMF approaches, as depicted in Fig. 8(b), (c), (f), and (i). The results generated by the Bayesian methods are very blurry, which indicates the loss of spatial information. As shown in Fig. 8(j) and (k), the noticeable geometric and spectral distortion of the red square region caused by the DDLPS and DRCNN methods can be found clearly. In comparison, the MRA and proposed methods are able to deliver a better visual quality of the fusion results for the Pavia Center data set, which is roughly consistent with the results of quantitative assessments listed in Table V. Moreover, the proposed DHP-DARN method can recover more accurate textures, such as the roof edges of the enlarged area in Fig. 8(l).

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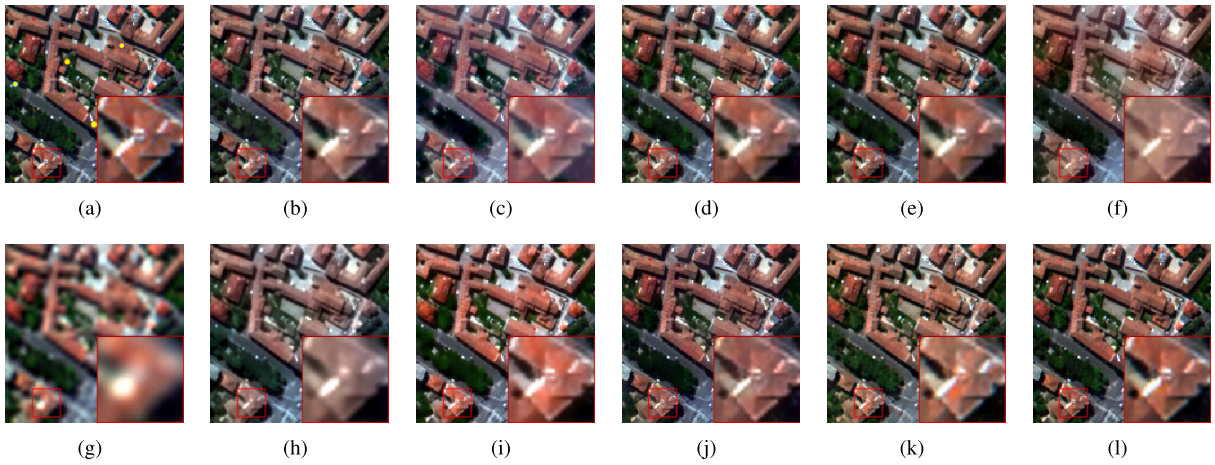



Fig. 8. Visual results generated by different algorithms for the *PaviaCenter_Patch21* (the 21st cubic-patch of the Pavia Center data set). (a) Reference. (b) GSA. (c) PCA. (d) SFIM. (e) MGH. (f) GFPCA. (g) BNA. (h) BSP. (i) CNMF. (j) DDLPS. (k) DRCNN. (l) Proposed.

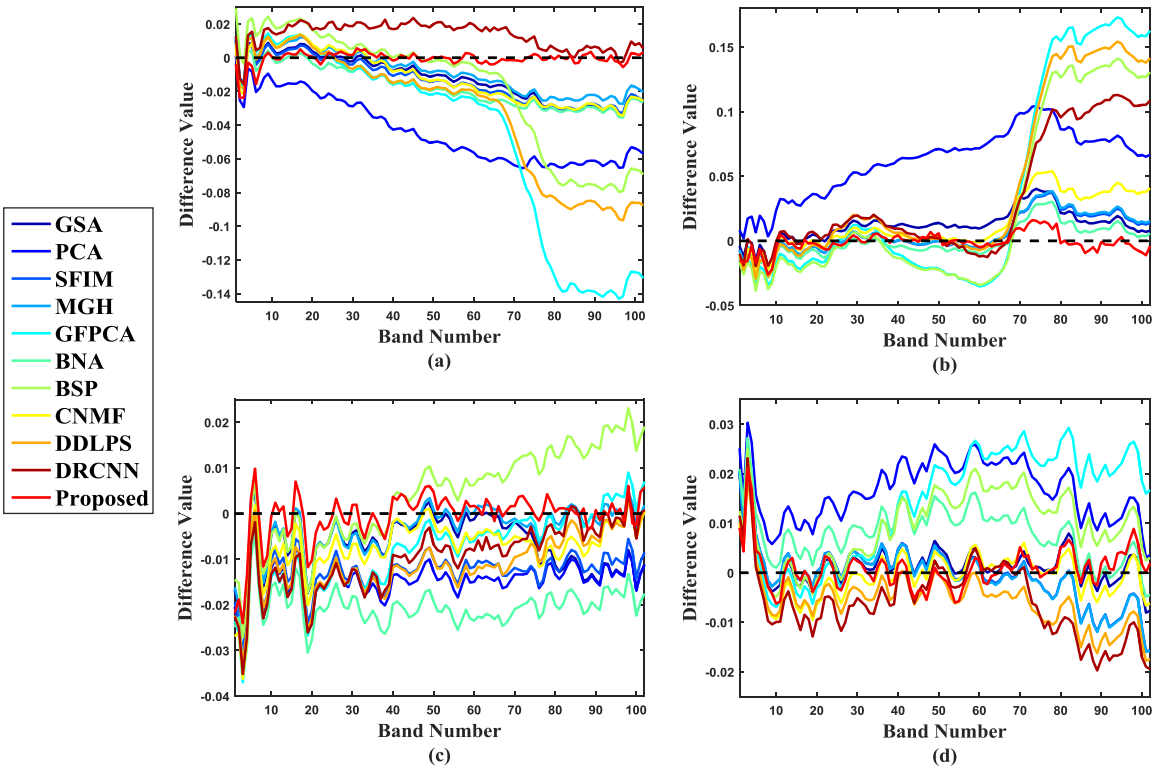



Fig. 9. Comparison of spectral reflectance difference values of four representative locations marked in Fig. 8(a). (a) Pixel (10,72). (b) Pixel (56,52). (c) Pixel (80,108). (d) Pixel (105,38).

In order to evaluate the spectral preservation capability of different HS pansharpening methods, Fig. 9 shows the spectral reflectance difference value comparison of four representative locations with coordinates (10,72), (56,52), (80,108), and (105,38), which are marked in yellow in Fig. 8(a). As depicted in Fig. 9, there is a dark dotted line serving as the benchmark in each subfigure, which means that being closer to the benchmark line indicates less spectral distortion. From Fig. 9, it can be obviously observed that the proposed DHP-DARN method obtains the best approximation of the reference HSI in the spectral domain, which demonstrates the superior spectral fidelity preservation of our proposal.

*3) Botswana Data Set:* For the Botswana data set, we first present the objective evaluation results in Table VI, where the average quantitative results of all the competing HS pansharpening methods on the testing set from the Botswana data set are listed. As can be seen from Table VI, the proposed method again obtains the best values of all the assessment metrics, which surpasses both the traditional and recent CNN-based HS pansharpening approaches with all the quantitative indexes ranking the first.

In addition, to perform the band-dependent quality evaluations of the fused HSIs for the Botswana data set, the average CC and PSNR curves as functions of the

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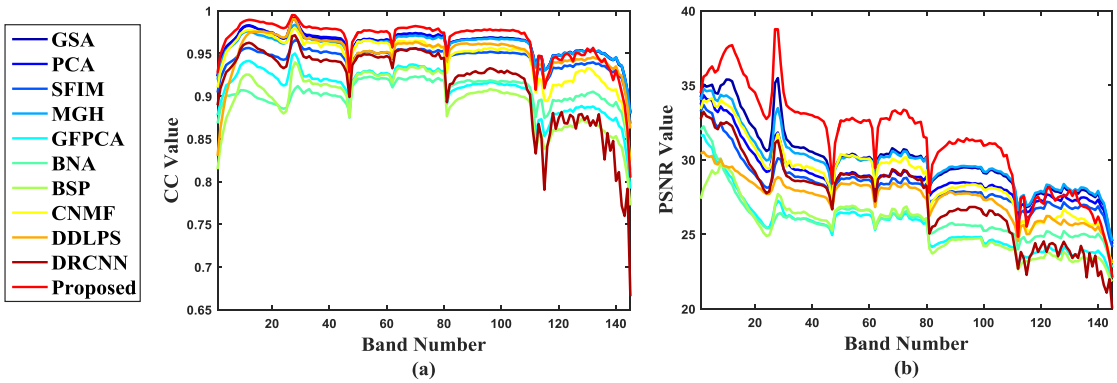



Fig. 10. Average CC and PSNR curves of the Botswana data set as functions of the spectral bands for different methods. (a) CC value. (b) PSNR value.

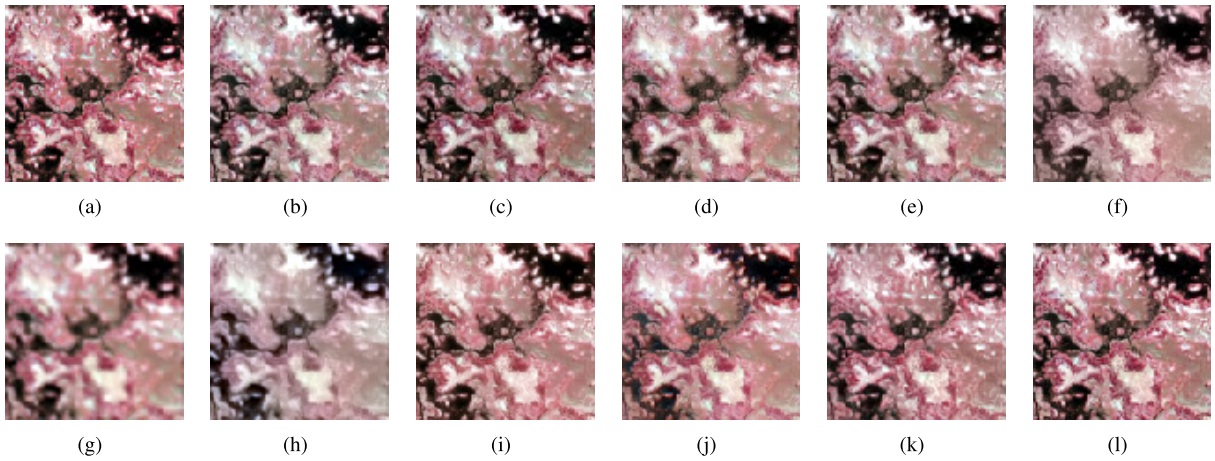



Fig. 11. Visual results generated by different approaches for the *Botswana_Patch15* (the 15th cubic-patch of the Botswana data set). (a) Reference. (b) GSA. (c) PCA. (d) SFIM. (e) MGH. (f) GFPCA. (g) BNA. (h) BSP. (i) CNMF. (j) DDLPS. (k) DRCNN. (l) Proposed.

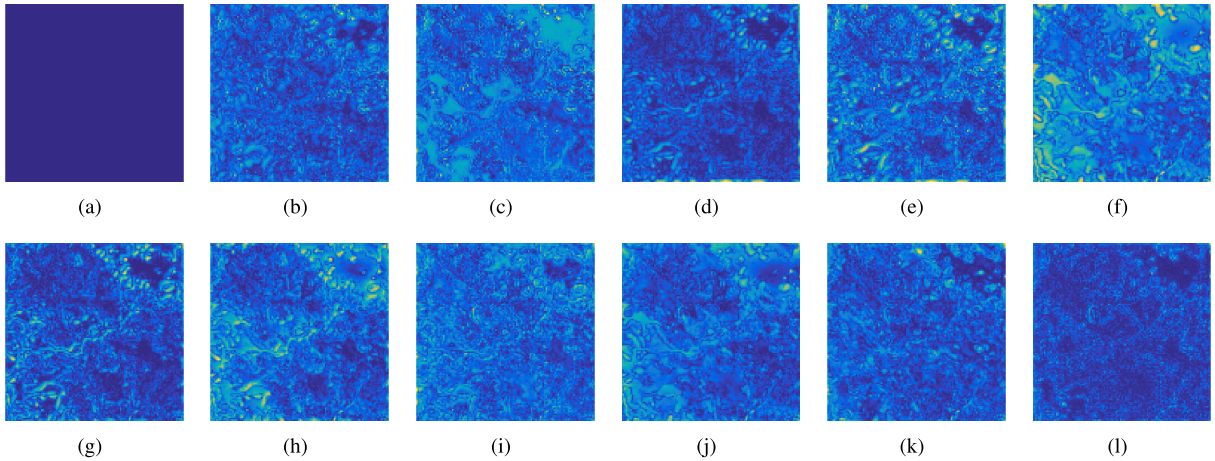



Fig. 12. Mean absolute error images of different approaches for the *Botswana_Patch15* (the 15th cubic-patch of the Botswana data set). (a) Reference. (b) GSA. (c) PCA. (d) SFIM. (e) MGH. (f) GFPCA. (g) BNA. (h) BSP. (i) CNMF. (j) DDLPS. (k) DRCNN. (l) Proposed.

spectral bands for different methods are depicted in Fig. 10. According to Fig. 10, the proposed method achieves the best CC and PSNR values among all the test methods for most spectral bands, which validates the higher spatial reconstruction quality of our DHP-DARN method once again.

For visual comparison of the Botswana data set, we present the fused HSIs of different approaches for the test image *Botswana_Patch15* (the 15th cubic-patch of the Botswana data set) in Fig. 11(b)–(l), and Fig. 11(a) shows the corresponding reference HR-HSI. Compared with the reference image, the CS, hybrid, and Bayesian methods all suffer from

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TABLE VII
QUANTITATIVE RESULTS OF NO-REFERENCE METRICS FOR DIFFERENT METHODS ON THE LOS ANGELES DATA SET

| Metric | GSA | PCA | SFIM | MGH | GFPCA | CNMF | DDLPS | DRCNN | Proposed |
|---|---|---|---|---|---|---|---|---|---|
| $D_\lambda$ ($\rightarrow$0) | 0.0750 | 0.0873 | 0.0650 | 0.0481 | 0.0697 | 0.0930 | 0.0529 | 0.0587 | **0.0459** |
| $D_S$ ($\rightarrow$0) | 0.0942 | 0.1326 | 0.0492 | 0.0700 | 0.1891 | 0.1692 | 0.1082 | 0.1080 | **0.0372** |
| QNR ($\rightarrow$1) | 0.8378 | 0.7916 | 0.8890 | 0.8853 | 0.7544 | 0.7535 | 0.8446 | 0.8397 | **0.9186** |

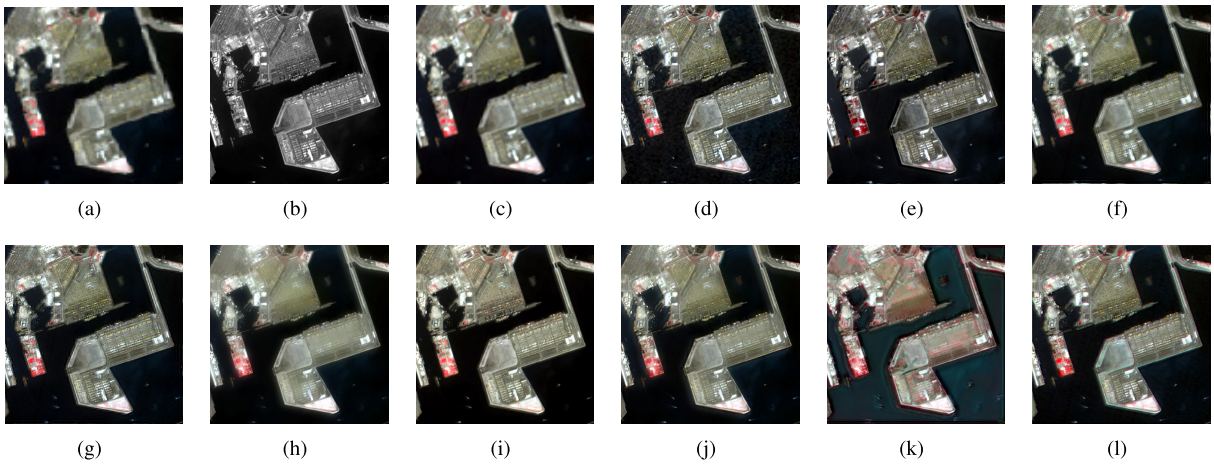



Fig. 13. Visual results obtained by different methods for the Los Angeles data set. (a) Real LR-HSI. (b) Real PAN image. (c) Upsampled HSI. (d) GSA. (e) PCA. (f) SFIM. (g) MGH. (h) GFPCA. (i) CNMF. (j) DDLPS. (k) DRCNN. (l) Proposed. Note that the LR-HSI is zoomed in for visualization.

serious spectral loss, which can be seen from the obvious color differences of their fusion results. By contrast, the MRA and CNMF approaches can yield relatively satisfactory results. Among the three CNN-based HS pansharpening algorithms, the proposed method performs the best in visual inspection, of which the fused result is the closest to the reference one.

To further verify the pansharpening performance of the proposed method for the Botswana data set, similar to the previous experiments, we display the mean absolute error images between the fused HSIs and the reference image *Botswana_Patch15* in Fig. 12, where bright yellow indicates large differences and dark blue represents small differences. In general, the error images in Fig. 12 are correspondent to the visual results in Fig. 11. It can be clearly seen from Fig. 12 that the proposed method can generate the fused result with the smallest error, which demonstrates the excellent fusion performance of our proposal.

### *E. Experiments on Real Hyperspectral Data Set*

In this section, to evaluate the generalization capability of the proposed method, experiments are conducted on the Los Angeles data set, which is a real HS data set. The two Bayesian approaches (BNA [15] and BSP [16]) used for performance comparison on simulated HS data sets are excluded in this section since the blur matrix is necessary during their implementation procedure, which is unavailable for the full-resolution validation of the Los Angeles data set.

To perform the fusion quality evaluation of this real HS data set without requiring a reference HR-HSI, we introduce the $D_\lambda$, $D_S$, and QNR [52] indexes, as described in Section IV-B. For objective assessment, the quantitative results of no-reference metrics for different methods on the Los Angeles data set are listed in Table VII. For visual inspection, the visual results obtained by different methods for the Los Angeles data set are depicted in Fig. 13. By comparison, the experimental results shown in Table VII and Fig. 13 fully demonstrate the superiority of the proposed method in terms of both quantitative indexes and visual appearance, which illustrates the high potential and generalization capability of our DHP-DARN method for HS pansharpening.

### *F. Ablation Experiments on CSA ResBlock and Residual Learning*

In this section, we demonstrate the effectiveness of the proposed CSA ResBlock and the residual learning strategy adopted in the DARN through ablation experiments on the Pavia Center data set.

*1) Effectiveness of the CSA ResBlock:* For the ablation study on the CSA ResBlock, besides the CSA ResBlock, we construct another three residual blocks for comparison, which are described as follows.

1) The base ResBlock contains two $3 \times 3$ Conv layers connected by one ReLU activation function, as shown in Fig. 14(a). Compared with the CSA ResBlock, the base ResBlock removes both the CA module and the SA module, corresponding to the first row in Table VIII.
2) The CA ResBlock is constructed by integrating only the CA module into the ResBlock for adaptively rescaling channelwise features, which is depicted in Fig. 14(b) and corresponds to the second row in Table VIII.
3) As illustrated in Fig. 14(c), the SA ResBlock is built by incorporating only the SA module into the ResBlock to

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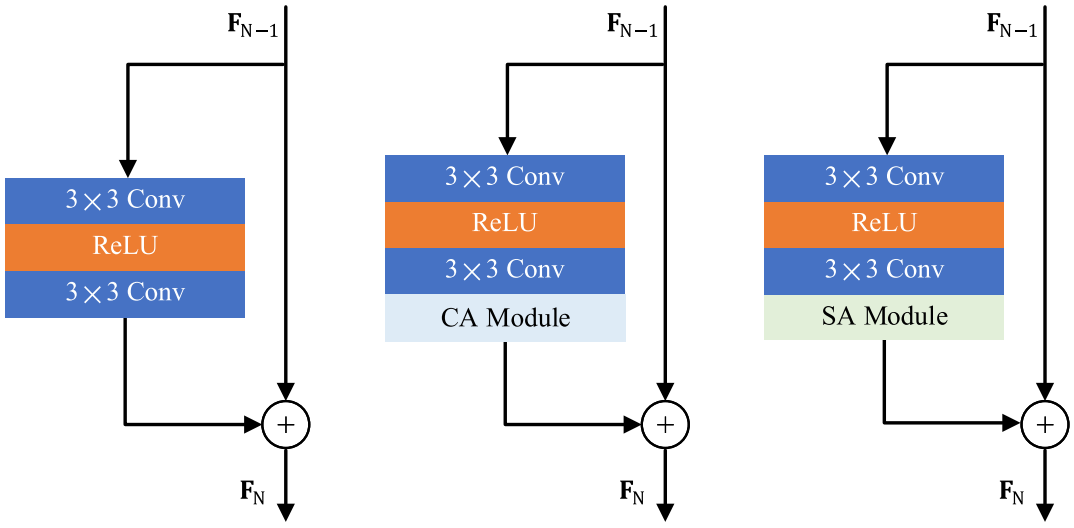



Fig. 14. Another three ResBlocks for comparison with our CSA ResBlock. (a) Base ResBlock. (b) CA ResBlock. (c) SA ResBlock.

recalibrate features spatially, which is corresponding to the third row in Table VIII.

In the ablation experiments, the number of ResBlocks is set to 4 for each competing network. For comparison, we present the quantitative fusion performance evaluations of the four networks with different ResBlock designs on the Pavia Center data set in Table VIII. It can be obviously observed from Table VIII that when both the CA module and the SA module are removed in the base ResBlock, the fusion performance is relatively low. Compared with the baseline model, all the quantitative results are apparently improved by integrating the CA or SA module into the ResBlock, especially when introducing the SA module into the unit. This is because the SA mechanism can highlight the texture information to enhance the contrast of the feature map, which benefits the reconstruction of high-frequency details during the HS pansharpening process. Moreover, when our proposed CSA ResBlock with the combinational complementary attention modules is utilized, the fusion performance can be further boosted, which demonstrates the superiority of our CSA ResBlock over the other ResBlock designs without attention module or with only one independent type of attention modules.

*2) Effectiveness of the Residual Learning:* To further improve the pansharpening accuracy, instead of learning a function that captures the relationship of the input $\mathbf{H}_{\text{in}}$ and the reference HR-HSI $\mathbf{H}_{\text{ref}}$ in a straightforward manner, we map the $\mathbf{H}_{\text{in}}$ to the residual HSI $\mathbf{H}_{\text{res}}$, which is the residual image between the $\mathbf{H}_{\text{ref}}$ and the upsampled HSI $\mathbf{H}_{\text{up}}$. In order to demonstrate the effectiveness of the residual learning, we perform the experiment on the Pavia Center data set using the nonresidual learning strategy, which directly learns a mapping from the input to the reference HSI. The training losses of the nonresidual learning method (denoted by proposed*) and the proposed method with residual learning on the Pavia Center data set are exhibited in Fig. 15. Obviously, the proposed method has a smaller loss and converges more quickly than the proposed* method. In addition, the average quantitative assessments of the proposed* method and the proposed method on the Pavia Center data set are listed in Table IX. As can be seen from Table IX, our proposal obtains better quality results than that of the proposed* method in terms of all the objective evaluation metrics. The results shown in Fig. 15 and Table IX fully verify that the residual learning strategy is really effective for accelerating the convergence speed and boosting the fusion performance.

TABLE VIII
AVERAGE QUANTITATIVE RESULTS OF DIFFERENT RESBLOCK DESIGNS ON THE PAVIA CENTER DATA SET

| ResBlock | CC (↑) | SAM (↓) | RMSE (↓) | ERGAS (↓) | PSNR (↑) |
|---|---|---|---|---|---|
| Base | 0.9752 | 4.5894 | 0.0190 | 2.9133 | 36.2582 |
| CA | 0.9784 | 4.2733 | 0.0173 | 2.7367 | 37.0054 |
| SA | 0.9799 | 4.1018 | 0.0164 | 2.6026 | 37.4864 |
| CSA | **0.9804** | **4.0909** | **0.0161** | **2.5771** | **37.6089** |

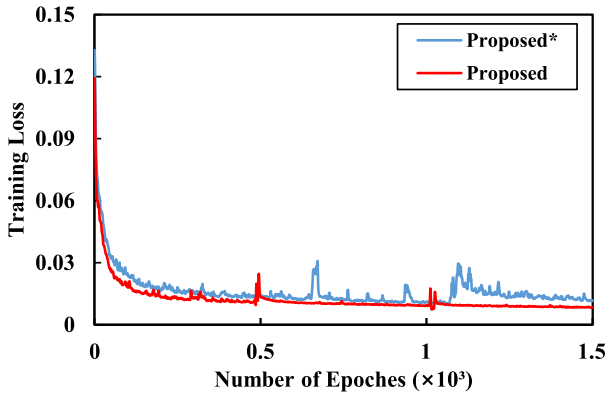



Fig. 15. Training losses of nonresidual learning and residual learning (proposed* and proposed) methods on the Pavia Center data set.

## V. CONCLUSION

In this article, we have presented a new framework based on DHP and DARN for HS pansharpening, which mainly consists of three parts: upsampling the LR-HSI via DHP,

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TABLE IX
AVERAGE QUANTITATIVE RESULTS OF NONRESIDUAL LEARNING AND RESIDUAL LEARNING (PROPOSED* AND PROPOSED) METHODS ON THE PAVIA CENTER DATA SET

| Method | CC (↑) | SAM (↓) | RMSE (↓) | ERGAS (↓) | PSNR (↑) |
|---|---|---|---|---|---|
| Proposed* | 0.9792 | 4.2814 | 0.0168 | 2.6877 | 36.9168 |
| Proposed | **0.9804** | **4.0909** | **0.0161** | **2.5771** | **37.6089** |

mapping the residual HSI through DARN, and obtaining the fused HSI by summation operation. To better preserve both spatial and spectral information, we first exploit the DHP algorithm to upsample the LR-HSI without learning from large example images. Then, the concatenated input of upsampled HSI and HR-PAN image is fed into the DARN with four CSA ResBlocks to map the residual image between the reference HR-HSI and the upsampled HSI. The CSA ResBlock is constructed by incorporating the CA module and the SA module into the ResBlock to adaptively excite more informative features and suppress the opposite ones across spectral channels and spatial positions simultaneously, which can effectively achieve outstanding performance for the fusion process. Finally, the fused HSI is obtained by the summation of the reconstructed residual HSI and the upsampled HSI. Comprehensive experiments on both simulated and real HS data sets demonstrate the superiority of our newly proposed DHP-DARN method for HS pansharpening in terms of quantitative and visual evaluations.

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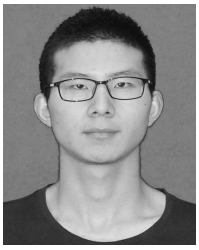
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